An Elementary Finite-Jet Approach to Sign Laws on Hyperplane Arrangements

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Abstract

We give an elementary finite-jet treatment of sign laws prescribed on whole orthants and on chambers of finite real affine hyperplane arrangements. A first nonzero homogeneous Taylor term inherits the prescribed signs weakly; polynomial factor multiplicities then force a signed parity law. A local-overlap argument propagates the crossing parity along each hyperplane. The resulting C^∞ obstruction is an instance of the established signature-flatness theory of Broglia and Tognoli, which also contains a leading-Taylor-term parity argument in a special crossing example. The contribution here is a systematic elementary arrangement-level proof, explicit finite-regularity forms (including a C^N arrangement version), and a synthesis with ultradifferentiable realization.

In compactwise Roumieu Denjoy–Carleman classes, the classical quasianalyticity criterion and positive flat functions yield a dichotomy: precisely affine Boolean laws on whole orthants in the quasianalytic case, and all Boolean laws in the non-quasianalytic case. For m distinct hyperplanes with r chambers, the corresponding counts are 2^{m+1} and 2^r. We place these statements in the older literature on distributions of signs and signatures, and in the Varchenko–Gelfand tradition of chamber functions. The emphasis is proof economy, normalization, and a transparent bridge among finite jets, sign rigidity, and ultradifferentiable realizability rather than historical firstness of the smooth obstruction.

1 Introduction

A Boolean truth table is finite. Requiring the same truth value throughout an entire orthant of $\mathbb{R}^n$ is not. This distinction becomes sharp when the Boolean value is extracted from the sign of a smooth real-valued function.

For example, the polynomial

$$F(x, y) = xy$$

is positive in the first and third quadrants and negative in the other two. With positive interpreted as 1 and negative as 0, it realizes equivalence (XNOR), an affine Boolean function over $\mathbb{F}_2$. By contrast, the AND sign pattern is positive only in the positive-positive quadrant and negative in the other three. The nonsmooth function $\min(x, y)$ has exactly that sign pattern, but the results below show that any *smooth* realization of AND on all four quadrants must be flat at the origin. In a quasianalytic class this is impossible for a nonzero function; in a non-quasianalytic class it is possible by using flat one-sided selectors.

The first purpose of this paper is to isolate the local mechanism behind that phenomenon. It is not coordinate division, resolution of singularities, or a special feature of one ultradifferentiable class. It is the first nonzero homogeneous Taylor term. If a smooth function has a nonzero finite jet at the common corner, its leading polynomial already inherits the orthant signs in a weak sense. Polynomial factorization across coordinate hyperplanes then forces an affine parity law.

The last factor-multiplicity step is elementary and has clear antecedents; the point of the finite-jet formulation is that it transports that obstruction from polynomials to arbitrary smooth germs and makes the subsequent regularity classification immediate.

The second purpose is to carry that mechanism to a finite affine hyperplane arrangement. If a smooth function is nonzero off the arrangement and has no flat point on it, the local leading polynomial at each point fixes a crossing parity for every incident hyperplane. Those parities propagate along each connected affine hyperplane and hence determine the sign on every chamber.

The third purpose is to combine these smooth statements with the classical Denjoy–Carleman theorem. We use throughout one compactwise Roumieu convention,

$$|D^\alpha f(x)| \le C_K \rho_K^{|\alpha|} |\alpha|! M_{|\alpha|}, \qquad x \in K \Subset U,$$

and no derivative-stability assumption. In this normalization the quasianalytic criterion is

$$\sum_{k=0}^{\infty} \frac{M_k}{(k+1)M_{k+1}} = \infty.$$

On the divergent side, quasianalyticity rules out the flatness required by a non-affine orthant law. On the convergent side, a classical compactly supported cutoff yields a one-sided selector, from which arbitrary Boolean laws are assembled without cancellation.

Prior work and contribution boundary. The smooth arrangement conclusion below is a consequence of the signature-flatness theorem of Broglia and Tognoli [13, Theorem 4.2]. Indeed, given F nonzero off the arrangement, put

$$G = F \prod_{j=1}^{m} \ell_j^2.$$

Then G has exactly the arrangement as its zero set and the same chamber signs. Wherever F has a first nonzero Taylor term, the first term of G is its product with a nonzero polynomial, so nonflatness is preserved. For $n \ge 2$, the homological hypothesis in their theorem is automatic on $\mathbb{R}^n$ because $H_{n-1}(\mathbb{R}^n; \mathbb{Z}_2) = 0$; the hyperplane components have global affine equations and are arc-analytically connected. The one-dimensional case is elementary and is covered directly below. For hyperplanes, their admissible signatures [13, Definition 2.2] are precisely the signed parity laws used here. Their normal-crossing refinement [13, Remark 4.6] also gives the smooth orthant flatness obstruction. Moreover, their discussion already uses a first nonzero Taylor coefficient to force an even/odd multiplicity contradiction in a crossing example. Thus neither the C^∞ obstruction nor the leading-term idea itself is claimed as new; the aim here is to package that mechanism systematically for affine arrangements and finite regularity.

This result belongs to a broader real-algebraic and real-analytic literature on *distributions of signs* and *signatures*. Fortuna explicitly studies sign distributions on connected components of complements of codimension-one real algebraic sets [5]. Acquistapace and Broglia develop the relation between signatures and flatness [1, 2]; subsequent work treats local completion and approximation questions [4, 3]. These papers have broader algebraic and analytic aims than the elementary arrangement argument used here, but they are the natural historical home for the sign-rigidity phenomenon.

On the arrangement side, locally constant functions on chambers and the Heaviside functions of the defining half-spaces are classical objects of Varchenko and Gelfand [7]; the resulting filtration

and its extensions to affine arrangements and oriented matroids are developed further, for example, by Moseley [8]. The signed parity laws below form only a very small multiplicative family inside the full algebra of chamber functions. Our question is not which chamber functions exist abstractly, but which of their $\{\pm 1\}$-valued sign patterns can arise from a single function subject to smoothness or ultradifferentiable constraints.

The Denjoy–Carleman criterion, cutoff existence, and stronger positive-flat constructions are classical [16, 14, 17]. The polynomial weak-sign argument is elementary and is not presented as new: Broglia–Tognoli's factorization analysis already records how parity of factor multiplicities controls a polynomial-induced signature [13, Proposition 2.4]. A later MathOverflow discussion gives the same observation in coordinate-orthant language [15]; finite-grid polynomial sign representation is a different established problem [12]. Stronger division, resolution and monomialization frameworks provide context [9, 11, 10], but are not required by the direct proofs below.

Accordingly, the paper is intended as a focused analysis note. Its contribution is not the first-nonzero-Taylor-term idea itself, but its systematic arrangement-level packaging, the explicit finite-order formulations, and the normalized synthesis with ultradifferentiable realization and exact orthant/chamber counts. No historical firstness is asserted for the underlying smooth obstruction, the existence of ultradifferentiable cutoffs or positive-flat functions, or the chamber-function formalism.

Organization. Section 2 fixes the sign and Denjoy–Carleman conventions. Section 3 proves the weak-sign polynomial lemma. Section 4 establishes finite-jet rigidity and the smooth arrangement theorem. Section 5 constructs one-sided selectors and proves the whole-orthant dichotomy. Section 6 gives the arrangement classification and exact counts. Section 7 records counterexamples that delimit the statements and clarifies the relation with normal-crossing results.

2 Definitions, conventions, and classical input

2.1 Orthants and Boolean sign laws

For $\alpha = (\alpha_1, \dots, \alpha_n) \in \{0,1\}^n$, put

$$\varepsilon_i(\alpha) = 2\alpha_i - 1 \in \{-1, +1\}$$

and define the open orthant

$$\mathcal{O}_\alpha = \{x \in \mathbb{R}^n : \varepsilon_i(\alpha) x_i > 0 \text{ for every } i\}.$$

A function F defined on an open set U meeting all these orthants *realizes* a Boolean law $\Theta : \{0,1\}^n \to \{0,1\}$ on the orthants in U if F is nonzero on their intersections with U and

$$\operatorname{sgn} F(x) = 2\Theta(\alpha) - 1, \qquad x \in \mathcal{O}_\alpha \cap U.$$

Local realization at the common corner means realization on a centered ball contained in U; global realization means $U = \mathbb{R}^n$. We call Θ *affine* if

$$\Theta(\alpha) = b \oplus e_1\alpha_1 \oplus \cdots \oplus e_n\alpha_n, \qquad b, e_i \in \{0,1\},$$

where $\oplus$ denotes addition in $\mathbb{F}_2$. There are 2^{n+1} affine Boolean functions. When $n = 1$, these are all four unary Boolean functions.

A smooth function is *flat at* p if $D^\alpha F(p) = 0$ for every multi-index α.

Definition 2.1 (Taylor-quasianalytic class)**.** For the two generic corollaries below, a smooth function class is called *Taylor-quasianalytic* if, at every point of its domain, a function whose full Taylor jet vanishes has zero germ there. For a vector space of germs, this zero-jet implication is equivalent to injectivity of the Taylor map. Only the zero-jet implication is used in the two generic corollaries below. No division, composition, or sheaf axiom is intended by this shorthand. The divergent Denjoy–Carleman classes used later have this property by the Denjoy–Carleman theorem.

2.2 Affine arrangements

Let

$$\mathcal{A} = \{H_1, \dots, H_m\}, \qquad H_j = \{x \in \mathbb{R}^n : \ell_j(x) = 0\},$$

be a finite family of distinct proper affine hyperplanes, where each nonconstant affine defining form ℓ_j is fixed once and for all. A *chamber* is a connected component of

$$\mathbb{R}^n \setminus \bigcup_{j=1}^{m} H_j.$$

We write $\mathrm{Ch}(\mathcal{A})$ for the chamber set. On a chamber C, every $\operatorname{sgn} \ell_j$ is constant. A chamber labeling $h : \mathrm{Ch}(\mathcal{A}) \to \{\pm 1\}$ is a *signed parity law* if

$$h(C) = c \prod_{j=1}^{m} \operatorname{sgn}(\ell_j|_C)^{e_j}, \qquad c \in \{\pm 1\}, \quad e_j \in \{0, 1\}. \tag{1}$$

The word "signed" is deliberate: when $c = -1$, this is not literally a group character, and the realized chamber sign vectors need not form a group.

2.3 One Denjoy–Carleman convention

Throughout Sections 5–6, a *normalized weight* means a sequence $M = (M_k)_{k \geq 0}$ such that

$$M_0 = 1, \qquad M_k > 0,$$

with M nondecreasing and log-convex. For an open set $U \subseteq \mathbb{R}^n$, define the compactwise Roumieu class $\mathcal{E}^{\{M\}}(U)$ by requiring that for every compact $K \Subset U$ there exist $C_K, \rho_K > 0$ such that

$$|D^\alpha f(x)| \leq C_K \rho_K^{|\alpha|} |\alpha|! M_{|\alpha|} \qquad (x \in K). \tag{2}$$

This convention keeps the factorial outside M. To compare with the common total-weight convention, set

$$W_k = k! M_k.$$

Then W is a weight sequence in the sense used in [16]: it is positive and log-convex, $W_0 = 1 \leq W_1$, and $W_k^{1/k} \to \infty$. Its quotient sequence satisfies

$$\frac{W_{k+1}}{W_k} = (k+1) \frac{M_{k+1}}{M_k},$$

so the Denjoy–Carleman series is precisely

$$\sum_{k=0}^{\infty} \frac{M_k}{(k+1)M_{k+1}}. \tag{3}$$

The Denjoy–Carleman theorem says that $\mathcal{E}^{\{M\}}$ is quasianalytic exactly when (3) diverges; see [16, Theorem 3.6]. In the convergent case, nontrivial compactly supported functions exist; the cutoff form we use is supplied, for example, by [16, Corollary 3.12].

We will need only elementary closure properties in addition to this classical input.

Lemma 2.2 (Elementary closure)**.** *For a normalized weight M, the class $\mathcal{E}^{\{M\}}$ is closed under finite sums, finite products, translations, and precomposition with affine linear functionals. For functions on $\mathbb{R}$ it is also closed under the primitive $G(t) = \int_0^t g(s)\,ds$.*

Proof. Only products and primitives need comment. Log-convexity with $M_0 = 1$ implies $M_j M_{k-j} \le M_k$. Combining this with Leibniz' rule and the bounds (2) gives the same form of estimate with a larger radius constant.

For a one-variable $g \in \mathcal{E}^{\{M\}}(\mathbb{R})$, define $G(t) = \int_0^t g(s)\,ds$. On a compact interval, G itself is bounded. For $k \ge 1$, $G^{(k)} = g^{(k-1)}$, and monotonicity of M gives

$$|G^{(k)}| \le C\rho^{k-1}(k-1)!M_{k-1} \le C'(\rho')^k k!M_k$$

for suitable enlarged constants. For an affine functional $\ell(x) = a \cdot x + b$ and a one-variable function g, the chain rule gives

$$D^\alpha(g \circ \ell)(x) = a^\alpha g^{(|\alpha|)}(\ell(x)).$$

The image of a compact set is compact, and the factors from a are absorbed into the radius constant. Translations are the one-variable special case of slope one. □

3 A weak-sign polynomial lemma

The finite-jet argument rests on an elementary fact: even a *weak* chamber sign prescription for a nonzero polynomial has parity form.

Lemma 3.1 (Polynomial weak-sign lemma)**.** *Let $\mathcal{A} = \{H_1, \dots, H_m\}$ be a finite family of distinct proper affine hyperplanes in $\mathbb{R}^n$, with fixed defining forms ℓ_j. Let P be a nonzero real polynomial. Suppose numbers $h_C \in \{\pm 1\}$ are assigned to the chambers $C \in \mathrm{Ch}(\mathcal{A})$ and satisfy*

$$h_C P(x) \ge 0 \qquad (x \in C). \tag{4}$$

Then there are unique $c \in \{\pm 1\}$ and $e_1, \dots, e_m \in \{0, 1\}$ such that

$$h_C = c \prod_{j=1}^{m} \operatorname{sgn}(\ell_j|_C)^{e_j} \qquad (C \in \mathrm{Ch}(\mathcal{A})). \tag{5}$$

The polynomial is allowed to vanish at points inside a chamber.

Proof. For each j, let a_j be the largest exponent for which $\ell_j^{a_j}$ divides P, and write

$$P = \ell_j^{a_j} Q_j, \qquad \ell_j \nmid Q_j.$$

The restriction of Q_j to H_j is therefore a nonzero polynomial on H_j and cannot vanish on a relatively open subset of H_j.

Consider two chambers sharing a facet contained in H_j. In the relative interior of that facet choose a point p outside every other hyperplane and with $Q_j(p) \neq 0$. In a sufficiently small ball about p, the sign of Q_j is fixed, while ℓ_j changes sign when H_j is crossed. Hence the sign of P changes by the factor $(-1)^{a_j}$. Wherever $P \neq 0$, the weak inequality (4) forces its sign to equal the prescribed chamber sign. Thus every adjacency across H_j has the same ratio

$$\frac{h_{C'}}{h_C} = (-1)^{a_j}.$$

Put $e_j \equiv a_j \pmod 2$.

Now divide the chamber label by $\prod_j \operatorname{sgn}(\ell_j)^{e_j}$. The quotient is unchanged across every chamber adjacency. The chamber adjacency graph is connected: choose interior points of two chambers and perturb a segment, or use a two-segment polygonal path, so that it meets the hyperplanes transversely and never meets two at the same time. The quotient is therefore a single constant $c \in \{\pm 1\}$, proving (5).

For uniqueness, fix j and choose a point of H_j outside all other hyperplanes. The two local chambers on its two sides differ only in the sign of ℓ_j, so their label ratio determines e_j. One chamber then determines c. □

Remark 3.2. The content of Lemma 3.1 is elementary polynomial factorization plus chamber adjacency. We use it as a transparent engine for the smooth results; we do not attach an independent novelty claim to it.

4 Finite-jet rigidity

4.1 Orthants

Theorem 4.1 (Finite-jet rigidity)**.** *Let F be C^∞ on a neighborhood of the origin in $\mathbb{R}^n$. Assume that in some ball B centered at the origin,*

$$F(x) \neq 0 \qquad \textit{whenever every coordinate of } x \textit{ is nonzero.}$$

Because each $B \cap \mathcal{O}_\alpha$ is connected, continuity makes the sign of F constant there; write the resulting local orthant sign law as $h : \{\pm 1\}^n \to \{\pm 1\}$. If at least one derivative of F at the origin is nonzero, then there are unique $c \in \{\pm 1\}$ and $e \in \{0,1\}^n$ such that

$$h(s) = c \prod_{i=1}^{n} s_i^{e_i}. \tag{6}$$

Equivalently, every smooth realization of a non-affine Boolean orthant law is flat at the origin.

Proof. Let d be the least order of a nonzero derivative of F at the origin; $d = 0$ is allowed. The first nonzero homogeneous Taylor term is

$$P_d(x) = \sum_{|\alpha|=d} \frac{D^\alpha F(0)}{\alpha!} x^\alpha, \qquad P_d \neq 0.$$

For any $v \in \mathbb{R}^n$ with no zero coordinate, Taylor's formula along the ray tv, $t \downarrow 0$, gives

$$F(tv) = t^d P_d(v) + o(t^d).$$

By the nonvanishing hypothesis and connectedness just noted, $F(tv)$ has sign $h(\operatorname{sgn} v)$ for every sufficiently small $t > 0$,

$$h(\operatorname{sgn} v)\, P_d(v) \geq 0.$$

Thus the nonzero polynomial P_d satisfies a weak sign prescription on the coordinate-hyperplane arrangement. Lemma 3.1 gives (6) and its uniqueness.

Under the identification $s_i = 2\alpha_i - 1$, a sign law of the form (6) is equivalent to an affine Boolean function $b \oplus e_1\alpha_1 \oplus \cdots \oplus e_n\alpha_n$. The contrapositive gives the flatness statement. □

Remark 4.2 (Finite regularity). The proof uses only the first nonzero Taylor order. If the first nonzero jet has order d, then a C^d Taylor expansion is enough. The theorem does not assert that nonflatness is necessary for an affine law: flat functions can also realize affine laws.

Corollary 4.3 (Finite-regularity form)**.** *Let $d \geq 0$ and let F be C^d near the origin, nonzero whenever all coordinates are nonzero in some centered ball. If every derivative of order $< d$ vanishes at the origin and at least one derivative of order d is nonzero, then the local orthant sign law is affine parity.*

Proof. The order-d Taylor expansion supplies the same nonzero leading homogeneous polynomial used in Theorem 4.1; Lemma 3.1 then applies verbatim. □

Corollary 4.4 (Quasianalytic orthant rigidity)**.** *Let F belong to a Taylor-quasianalytic class on a neighborhood of the origin and suppose it realizes a whole local orthant law there. Then that Boolean law is affine.*

Proof. If every derivative of F vanished at the origin, the defining zero-jet property would make the germ of F at the origin zero, contradicting nonvanishing on every local open orthant. Apply Theorem 4.1. □

4.2 Affine hyperplane arrangements

Theorem 4.5 (No-flat-point arrangement rigidity)**.** *Let $\mathcal{A} = \{H_1, \ldots, H_m\}$ be a finite family of distinct proper affine hyperplanes in $\mathbb{R}^n$. Let $F \in C^\infty(\mathbb{R}^n)$ be nonzero on $\mathbb{R}^n \setminus \bigcup_j H_j$. Assume that F is not flat at any point of $\bigcup_j H_j$. Then there are unique $c \in \{\pm 1\}$ and $e_1, \ldots, e_m \in \{0, 1\}$ such that*

$$\operatorname{sgn} F(x) = c \prod_{j=1}^{m} \operatorname{sgn} \ell_j(x)^{e_j}, \qquad x \notin \bigcup_j H_j. \tag{7}$$

Proof. Fix $p \in \bigcup_j H_j$. Choose a ball B_p small enough that every hyperplane meeting B_p contains p. After translating p to the origin, the incident hyperplanes form a central arrangement. Let P_p be the first nonzero homogeneous Taylor polynomial of F at p.

For every direction v outside the incident hyperplanes,

$$F(p + tv) = t^{d(p)} P_p(v) + o(t^{d(p)}), \qquad t \downarrow 0.$$

The function F has a fixed sign on each local chamber because it is continuous and nonzero there. Hence P_p inherits those signs weakly. Lemma 3.1 gives, for every hyperplane H_j through p, a local crossing parity $e_j(p) \in \{0, 1\}$.

We claim that $e_j(p)$ is locally constant as p varies in H_j. Fix $p \in H_j$ and its ball B_p. For $p' \in H_j$ sufficiently close to p, choose the local ball $B_{p'}$ contained in B_p and avoiding all hyperplanes not containing p'. The relatively open set $H_j \cap B_{p'}$ contains a point q outside every other hyperplane: the excluded intersections form a finite union of proper affine subspaces of H_j. In dimension one, H_j is a point and distinct hyperplanes do not meet it. A sufficiently small transverse pair near q, on opposite sides of H_j, lies inside both balls and meets no other hyperplane. The ratio of the two actual signs of F is both $(-1)^{e_j(p)}$ and $(-1)^{e_j(p')}$. Thus the bits agree. Since H_j is connected, $e_j(p)$ is a single global bit e_j.

Now consider the global chamber adjacency graph. Across a generic facet contained in H_j, the sign of F changes by $(-1)^{e_j}$. The same is true of the product $\prod_j \operatorname{sgn}(\ell_j)^{e_j}$. Their quotient is therefore unchanged across every chamber adjacency. As in Lemma 3.1, the chamber adjacency graph is connected, so the quotient is a global constant $c \in \{\pm 1\}$. This proves (7). Uniqueness follows by crossing a generic point of each H_j. □

Corollary 4.6 (Finite-regularity arrangement form)**.** *Let $N \geq 0$, let $\mathcal{A} = \{H_1, \ldots, H_m\}$ be a finite family of distinct proper affine hyperplanes in $\mathbb{R}^n$, and let $F \in C^N(\mathbb{R}^n)$ be nonzero on $\mathbb{R}^n \setminus \bigcup_j H_j$. Assume that for every $p \in \bigcup_j H_j$ the N-jet of F at p is nonzero; equivalently, for each such p there is a multi-index α with $|\alpha| \leq N$ and $D^\alpha F(p) \neq 0$. Then the chamber sign law of F is a unique signed parity law of the form* (7)*.*

Proof. At each p on the arrangement choose the least $d(p) \leq N$ for which some derivative of order $d(p)$ is nonzero. The $C^{d(p)}$ Taylor expansion gives a nonzero leading homogeneous polynomial and therefore the same local crossing parities as in the proof of Theorem 4.5. The propagation of those parities along a hyperplane, and then across the connected chamber adjacency graph, uses only continuity and nonvanishing off the arrangement. Thus the remainder of that proof applies unchanged. □

Remark 4.7 (A quantitative obstruction)*.* Equivalently, if a C^N function is nonzero off the arrangement and realizes a chamber labeling that is not signed parity, then at some point of the arrangement its entire N-jet must vanish. The smooth theorem is a stronger all-orders analogue: it yields a single flat point. The finite-order statements alone do not imply that conclusion by letting N tend to infinity, because the point at which the N-jet vanishes may depend on N.

Corollary 4.8 (Quasianalytic arrangement rigidity)**.** *Let F be a nonzero globally defined member of a Taylor-quasianalytic class on $\mathbb{R}^n$, and suppose F is nonzero off a finite affine arrangement. Then its chamber sign law has the form* (7)*.*

Proof. At a flat point, the defining zero-jet property would make the germ of F there zero. Every neighborhood of a point on a finite hyperplane arrangement contains points off the arrangement, contradicting the assumed nonvanishing there. Thus F has no flat point on the arrangement, and Theorem 4.5 applies. □

5 Denjoy–Carleman realization on orthants

The finite-jet theorem supplies the rigidity half of the classification. The converse direction in a non-quasianalytic class is constructive.

Lemma 5.1 (One-sided selector)**.** *Let M be a normalized weight. If*

$$\sum_{k=0}^{\infty} \frac{M_k}{(k+1)M_{k+1}} < \infty, \tag{8}$$

then there exists $\psi \in \mathcal{E}^{\{M\}}(\mathbb{R})$ such that

$$\psi(t) = 0 \quad (t \le 0), \qquad \psi(t) > 0 \quad (t > 0).$$

Proof. With $W_k = k!M_k$, condition (8) is the non-quasianalytic Denjoy–Carleman condition $\sum W_k/W_{k+1} < \infty$. By the classical cutoff theorem, choose a nonzero real-valued compactly supported function f in the corresponding Roumieu class; see [16, Theorem 3.6 and Corollary 3.12]. Translate f so that the left endpoint of its support is 0. Then

$$f(t) = 0 \quad (t \le 0),$$

and every interval $(0, t)$, $t > 0$, contains a point at which f is nonzero: otherwise f would vanish on a right-neighborhood of 0, contradicting that 0 is the left endpoint of $\operatorname{supp} f$.

By Lemma 2.2, f^2 and its primitive belong to the same class. Define

$$\psi(t) = \int_0^t f(s)^2 \, ds.$$

For $t \le 0$ the integral is zero. For $t > 0$, the interval $(0, t)$ contains a point where $f \neq 0$; continuity then gives an interval on which $f^2 > 0$, and therefore $\psi(t) > 0$. □

Classical constructive antecedent. A stronger existence statement is classical. Rainer's Remark 3.13, citing work of Langenbruch, states that in the non-quasianalytic setting one can construct an ultradifferentiable function that is flat on a prescribed closed set and positive (indeed real analytic) on its complement [16, Remark 3.13]; see also [14]. Taking the closed half-line $(-\infty, 0]$ gives a one-sided selector. More generally, taking the closed set to be the union of an arrangement and multiplying the positive-flat function on each chamber by an arbitrarily prescribed sign, then extending by zero on the arrangement, directly realizes every chamber labeling: flatness makes the chamberwise pieces glue with zero jet on the walls, and the Roumieu derivative bounds are unchanged in absolute value. We nevertheless retain the squared-cutoff selector and the finite chamber sum because they give an explicit elementary formula adapted to the defining affine forms.

Definition 5.2. For a normalized weight M, let $\mathcal{R}_M(n)$ be the set of Boolean functions $\Theta : \{0,1\}^n \to \{0,1\}$ realized by some globally defined $F \in \mathcal{E}^{\{M\}}(\mathbb{R}^n)$ that is nonzero whenever all coordinates are nonzero.

Theorem 5.3 (Whole-orthant classification)**.** *Let M be a normalized weight. Then*

$$\mathcal{R}_M(n) = \begin{cases} \{\textit{affine Boolean functions}\}, & \displaystyle\sum_{k=0}^{\infty} \frac{M_k}{(k+1)M_{k+1}} = \infty, \\ \{\textit{all Boolean functions on } \{0,1\}^n\}, & \displaystyle\sum_{k=0}^{\infty} \frac{M_k}{(k+1)M_{k+1}} < \infty. \end{cases}$$

Proof. Assume first that the series diverges. By the Denjoy–Carleman theorem, $\mathcal{E}^{\{M\}}$ is quasianalytic. If F realizes a whole-orthant law, then F is not the zero function in any neighborhood of the origin, so its Taylor jet at the origin cannot vanish identically. Theorem 4.1 implies that the law is affine. Conversely, every affine law is realized by the polynomial

$$F(x) = c \prod_{i=1}^{n} x_i^{e_i}$$

for suitable $c \in \{\pm 1\}$ and $e_i \in \{0, 1\}$, and every polynomial lies in the class.

Now assume the series converges and choose the selector ψ from Lemma 5.1. For $\alpha \in \{0,1\}^n$, define

$$\Psi_\alpha(x) = \prod_{i=1}^{n} \psi(\varepsilon_i(\alpha) x_i).$$

For an arbitrary Boolean function Θ, set

$$F_\Theta(x) = \sum_{\alpha \in \{0,1\}^n} (2\Theta(\alpha) - 1)\Psi_\alpha(x). \tag{9}$$

By Lemma 2.2, $F_\Theta \in \mathcal{E}^{\{M\}}(\mathbb{R}^n)$. If $x \in \mathcal{O}_\alpha$, then $\Psi_\alpha(x) > 0$, while for every $\beta \neq \alpha$ at least one factor of $\Psi_\beta(x)$ is zero. Hence

$$\operatorname{sgn} F_\Theta(x) = 2\Theta(\alpha) - 1.$$

Thus every Boolean function is realized. □

Corollary 5.4 (Gevrey dichotomy)**.** *For the Gevrey class G^s, $s \geq 1$, with derivative growth $C\rho^k(k!)^s$, the whole-orthant laws are exactly the affine Boolean functions when $s = 1$, and all Boolean functions when $s > 1$.*

Proof. In the normalization (2), take $M_k = (k!)^{s-1}$. Then

$$\frac{M_k}{(k+1)M_{k+1}} = (k+1)^{-s}.$$

The corresponding series diverges exactly at $s = 1$ for $s \geq 1$. Apply Theorem 5.3. □

Remark 5.5. For $n = 1$, the two sides of Theorem 5.3 contain the same four Boolean functions because every unary Boolean function is affine. The strict expressive difference begins at arity $n \geq 2$.

6 Affine arrangements: classification and exact counts

The orthant construction extends directly once orthants are replaced by the realized sign cells of a finite affine arrangement.

Theorem 6.1 (Arrangement classification)**.** *Let M be a normalized weight. Let $\mathcal{A}$ be an arrangement of m distinct proper affine hyperplanes in $\mathbb{R}^n$, and write $r = |\operatorname{Ch}(\mathcal{A})|$.*

If the series (3) *diverges, the chamber sign laws realized by globally defined $F \in \mathcal{E}^{\{M\}}(\mathbb{R}^n)$ that are nonzero off the arrangement are exactly the signed parity laws* (1)*; there are exactly 2^{m+1} such laws.*

If the series (3) *converges, every one of the 2^r chamber labelings is realizable by such a function.*

Proof. On the quasianalytic side, Corollary 4.8 gives the signed-parity form. Every choice of $(c, e_1, \dots, e_m)$ is realized by the polynomial

$$c \prod_{j=1}^{m} \ell_j(x)^{e_j}.$$

These 2^{m+1} restrictions are distinct. Indeed, for each H_j choose a generic point of H_j outside all other hyperplanes. The adjacent chamber pair at that point differs only in the sign of ℓ_j, so its label ratio recovers e_j; one chamber then recovers c.

On the non-quasianalytic side, let ψ be the selector from Lemma 5.1. For each chamber C, let

$$s_j(C) = \operatorname{sgn}(\ell_j|_C)$$

and define

$$\Psi_C(x) = \prod_{j=1}^{m} \psi(s_j(C)\ell_j(x)).$$

Distinct chambers have distinct sign vectors: a nonempty fixed sign cell is an intersection of open half-spaces and is therefore convex and connected. Hence if x lies in a chamber D, then $\Psi_D(x) > 0$ and $\Psi_C(x) = 0$ for every $C \neq D$.

Given any labeling $h : \operatorname{Ch}(\mathcal{A}) \to \{\pm 1\}$, set

$$F_h(x) = \sum_{C \in \operatorname{Ch}(\mathcal{A})} h(C)\Psi_C(x).$$

Elementary closure gives $F_h \in \mathcal{E}^{\{M\}}(\mathbb{R}^n)$, and on chamber D the sum reduces to $h(D)\Psi_D(x)$. Thus F_h has the prescribed sign and is nonzero off the arrangement. □

Example 6.2 (Three concurrent lines)**.** Take three distinct lines through the origin in $\mathbb{R}^2$, for example

$$x = 0, \qquad y = 0, \qquad x - y = 0.$$

There are $r = 6$ chambers. A full quasianalytic normalized Roumieu class $\mathcal{E}^{\{M\}}$ therefore realizes $2^{3+1} = 16$ signed-parity chamber laws, whereas a full non-quasianalytic such class realizes all $2^6 = 64$ labelings.

Remark 6.3 (When the counts coincide)*.* Each added distinct hyperplane splits at least one existing chamber, so $r \geq m + 1$. The numerical gap is strict exactly when $r > m + 1$. For m parallel hyperplanes one has $r = m + 1$, so the two counts coincide. Thus the quasianalytic/non-quasianalytic distinction is structural, but its cardinality shadow depends on the arrangement.

7 Scope, counterexamples, and relation to nearby theory

7.1 Real zero support is weaker than a normal-crossing divisor

If one already has a factorization

$$F = u \prod_i x_i^{a_i}$$

with a nonvanishing unit u, then the sign law is immediately the sign of u times a parity product of coordinate signs. Normal-crossing and monomialization theories therefore provide important

Table 1: The classification in the common notation of this paper. All exact realizability counts refer to the full normalized Roumieu classes $\mathcal{E}^{\{M\}}$.

Setting	Mechanism	Sign laws
Smooth germ with a nonzero jet at the common orthant corner	Leading homogeneous Taylor polynomial and Lemma 3.1	Affine parity only
C^N function on an arrangement with nonzero N-jet at every wall point	Local leading terms plus parity propagation	Signed parity only
Quasianalytic $\mathcal{E}^{\{M\}}$ on orthants	Taylor injectivity excludes the flatness required by Theorem 4.1	Exactly 2^{n+1} affine Boolean laws
Non-quasianalytic $\mathcal{E}^{\{M\}}$ on orthants	One-sided flat selector	All 2^{2^n} Boolean laws
Quasianalytic $\mathcal{E}^{\{M\}}$ on m distinct hyperplanes	No flat points plus Theorem 4.5	Exactly 2^{m+1} signed-parity laws
Non-quasianalytic $\mathcal{E}^{\{M\}}$ on an arrangement with r chambers	Chamber selectors	All 2^r chamber labelings

neighboring context; see [11, 10]. They should not, however, be substituted for the weaker hypothesis used in Theorems 4.5 and 6.1.

Indeed,

$$F(x, y) = xy(x^2 + y^2) \tag{10}$$

is real analytic and has real zero set exactly equal to the union of the two coordinate axes. Its sign off those axes is $\operatorname{sgn}(xy)$. But after extracting the generic first-order factors x and y, the residual factor $x^2 + y^2$ still vanishes at the origin and is not a unit. Thus set-theoretic real zero support on an arrangement does not imply a unit-times-monomial factorization in the original coordinates.

This is why the arrangement theorem above is stated and proved directly at the sign level. Resolution and monomialization results involve stronger divisor hypotheses and/or transformations of source and target; they remain relevant background, not a replacement for the present argument.

7.2 A common corner is essential for local orthant rigidity

The connectedness of a domain alone does not replace the common-corner hypothesis in Theorem 4.1. On the punctured plane $U = \mathbb{R}^2 \setminus \{0\}$, define

$$F(x, y) = x + y + \sqrt{x^2 + y^2}. \tag{11}$$

This function is real analytic on U. It is positive in the positive-positive quadrant and in both mixed-sign quadrants, while it is negative in the negative-negative quadrant. In a mixed-sign quadrant,

$$\sqrt{x^2 + y^2} > |x + y|$$

because $xy < 0$, whereas in the negative-negative quadrant

$$\sqrt{x^2 + y^2} < |x + y| = -x - y.$$

Thus (11) realizes OR on the four quadrants of the punctured plane. There is no contradiction with Theorem 4.1: the origin, at which the finite-jet obstruction would live, has been removed.

7.3 Smoothness alone does not imply rigidity

The selector construction in (9) gives C^∞ realizations of every Boolean law in the non-quasianalytic case. For a non-affine law, Theorem 4.1 says that such a realization must be flat at the origin; the construction is consistent with exactly that requirement. The mathematical boundary is therefore not "smooth versus nonsmooth," but rather the presence or absence of a nonzero finite jet at the relevant common boundary.

7.4 Sign distributions, polynomial signs, and chamber functions

The factor-parity mechanism in Lemma 3.1 is classical antecedent material, not the paper's novelty. In Broglia–Tognoli's Proposition 2.4, maximal powers of irreducible defining polynomials determine whether a polynomial-induced signature changes sign along each component; for hyperplanes this is exactly the parity-bit mechanism used here [13, Proposition 2.4]. A later 2014 MathOverflow discussion gives the same observation in coordinate-orthant language: a polynomial required to switch sign across one facet but not another facet of the same coordinate hyperplane would force the multiplicity of the corresponding coordinate factor to be both odd and even [15]. There is also a separate literature on polynomial sign representations of Boolean functions on finite grids; for example, Basu, Bhatnagar, Gopalan, and Lipton study degree–sparsity tradeoffs for sign representations of parity [12]. Those finite-grid questions are different from requiring a sign throughout every point of each open orthant.

The more direct historical language is that of sign distributions and signatures. Fortuna [5] considers sign assignments to components of complements of codimension-one algebraic sets. Broglia and Tognoli [13] connect nonadmissible signatures with flatness; Acquistapace and Broglia continue this line in *Signatures and flatness*, together with its published correction [1, 2]. The local-completion problem on algebraic surfaces is studied by Acquistapace, Broglia, and Fortuna [4], while Acquistapace and Broglia relate signatures to approximation by regular or analytic functions [3]. The present arrangement statements should therefore be read as an elementary specialization and finite-jet repackaging of this tradition, not as the origin of the sign-rigidity phenomenon.

There is also a distinct combinatorial/algebraic tradition in which all locally constant chamber functions are studied. Varchenko and Gelfand introduced the algebra generated by Heaviside functions of an arrangement [7]; Moseley gives a modern treatment of the associated filtration for affine arrangements and oriented matroids [8]. For standard sign-vector, chamber, and tope language see also [6]. Our chamber selectors on the non-quasianalytic side are compatible with this viewpoint, but the analytic question is narrower: smoothness and quasianalyticity select the 2^{m+1} signed parity laws from among the 2^r possible $\{\pm 1\}$-valued chamber functions.

7.5 Established machinery versus the sign classification

The background literature contains several stronger analytic frameworks. Coordinate division is part of standard axiomatic treatments of quasianalytic classes under suitable closure assumptions;

see [9]. Quasianalytic local rings and division phenomena have a substantial independent literature; see [17]. Resolution and monomialization produce monomial forms after controlled modifications; see [11, 10]. None of those ingredients is claimed here as new.

Conversely, the proofs of Theorems 4.1 and 4.5 deliberately avoid coordinate division and resolution. Their input is the first nonzero Taylor polynomial at a point. The Denjoy–Carleman classification then uses only the classical quasianalyticity criterion and cutoff existence. This separation gives a self-contained proof of consequences of the earlier signature-flatness and ultra-differentiable existence results. The failure of a residual-unit factorization does not prevent the squared-multiplier reduction to [13].

8 Conclusion

The finite-jet proof gives a short route to an established smooth sign obstruction and its affine-arrangement form. A nonzero leading Taylor term forces affine parity on orthants; local crossing parities propagate along each hyperplane. Its elementary nature separates this mechanism from stronger division and resolution frameworks.

Combined with the classical Denjoy–Carleman criterion and positive flat functions, the argument gives an explicit realization classification. For m distinct affine hyperplanes with r chambers, a full normalized Roumieu class realizes 2^{m+1} signed-parity laws in the quasianalytic regime and all 2^r chamber labelings in the non-quasianalytic regime.

The contribution of the note is therefore not the first-nonzero-Taylor-term idea itself, but its systematic arrangement-level packaging, the finite-regularity orthant and arrangement forms, the explicit selector formulas, and the organized Denjoy–Carleman synthesis with exact realizability counts. The surrounding literature on distributions of signs, signatures, positive-flat functions, and chamber-function algebras makes clear that neither the underlying C^∞ obstruction nor the existence mechanism for arbitrary non-quasianalytic chamber signs originates here. Real zero support need not imply original-coordinate unit factorization, but that distinction does not prevent the reduction to earlier signature-flatness results. The note is therefore positioned around transparency, finite-order formulation, and explicit synthesis rather than a claim of first discovery.

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